

Software Requirements Specification Document

for

File Converter System (CSV to ARFF and ARFF to CSV)

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Table of Contents

1	Introduction	1
1.1	Purpose	1
1.2	Scope	1
1.3	Definitions, Acronyms, and Abbreviations	1
1.4	References	2
1.5	Overview	2
2	Overall Description	2
2.1	Product Perspective	2
2.1.1	System Interfaces	2
2.1.2	User Interfaces	2
2.1.3	Hardware Interfaces	3
2.1.4	Software Interfaces	3
2.1.5	Communications Interfaces	3
2.1.6	Operations	3
2.2	Product Functions	3
2.3	User Characteristics	3
2.4	Constraints	4
2.5	Assumptions and Dependencies	4
3	Specific Requirements	4
3.1	Functional Requirements	4
3.2	Design Constraints	6
3.3	Non-Functional Requirements	7
3.3.1	Performance Requirements	7
3.3.2	Reliability Requirements	7
3.3.3	Availability Requirements	7
3.3.4	Security Requirements	7
3.3.5	Maintainability Requirements	8
3.3.6	Portability Requirements	8
3.3.7	Correctness Requirements	8
3.4	External Interface Requirements	8
3.4.1	User Interface Requirements	8
3.4.2	Hardware Interface Requirements	8
3.4.3	Software Interface Requirements	9
3.4.4	Communications Interface Requirements	9
4	Appendix A: Protocols and Standards	9
5	Appendix B: System Architecture	10

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for the **File Converter System**, a web application that performs bidirectional conversion between the Comma-Separated Values (CSV) format and the Attribute-Relation File Format (ARFF) used by the WEKA machine learning workbench. It is intended for the development team, project guide, evaluators, and quality assurance personnel.

The system provides an authenticated platform on which registered users can upload a dataset in either format, review the detected attribute schema, convert the dataset to the required target format, and download the result, while maintaining a per-user history of conversions.

1.2 Scope

The File Converter System is a web-based data-preparation application that enables:

- **For Registered Users:** Create an account, sign in securely, upload CSV or ARFF files, configure parsing options, review and override the inferred schema, convert in either direction, preview and download the output, and view conversion history.
- **For Administrators:** Manage user accounts, monitor conversion activity and logs, and configure limits such as maximum file size.

The system shall be built as a two-tier web application, with a React.js frontend and a Python REST backend communicating over HTTPS, keeping presentation and conversion logic separate. Outside the scope of this version: formats other than CSV and ARFF, statistical analysis or model building on uploaded data, and sparse-format ARFF.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
Admin	Administrator user with system management privileges
API	Application Programming Interface
ARFF	Attribute-Relation File Format; an ASCII format used by WEKA describing a list of instances sharing a set of attributes
Attribute	A named, typed column of a dataset
CSV	Comma-Separated Values; a plain-text format in which each line is a record and fields are separated by a delimiter
DBMS	Database Management System
Delimiter	The character separating fields within a CSV record
HTTP / HTTPS	HyperText Transfer Protocol / HyperText Transfer Protocol Secure
IEEE	Institute of Electrical and Electronics Engineers
Instance	A single data record, corresponding to one row of the dataset
JSON	JavaScript Object Notation
JWT	JSON Web Token, used for stateless session authentication
Nominal Attribute	An attribute whose value is drawn from a finite enumerated set
REST	Representational State Transfer
Schema	The ordered list of attribute names together with their data types
UI / UX	User Interface / User Experience
WEKA	Waikato Environment for Knowledge Analysis; the workbench defining the ARFF format

1.4 References

Reference	Version	Updated	Source
IEEE Std 830-1998, Recommended Practice for SRS	1998	1998	IEEE Computer Society
ARFF Format Specification	Current	2024	https://www.cs.waikato.ac.nz/ml/weka/arff.html
RFC 4180, Common Format and MIME Type for CSV Files	RFC 4180	Oct 2005	IETF
React.js Documentation	v19	Dec 2024	https://react.dev
FastAPI Documentation	v0.115	2025	https://fastapi.tiangolo.com
pandas Documentation	v2.2	2024	https://pandas.pydata.org/docs
PostgreSQL Documentation	Version 17	Oct 2024	https://postgresql.org/docs
JWT Standards	RFC 7519	May 2015	IETF

1.5 Overview

Section 2 describes the general factors affecting the product: its perspective, major functions, user classes, constraints, and assumptions. Section 3 contains the detailed requirement specifications, comprising functional requirements, design constraints, non-functional requirements, and external interface requirements. Sections 4 and 5 provide supporting information in the form of the protocols and standards used and the system architecture diagram, and Section 6 provides the index.

2. Overall Description

2.1 Product Perspective

The File Converter System is a new, self-contained web application acting as a data-preparation utility between general-purpose tabular sources, which commonly export CSV, and the WEKA workbench, which requires ARFF. It operates independently but integrates with a relational database for account and history storage and with an email service for verification and password recovery. Internally it converts through a format-independent representation consisting of a relation name, an ordered attribute schema, and a collection of instances, so that both directions share a single validated core.

2.1.1 System Interfaces

- A relational DBMS for user accounts, sessions, and conversion history
- An email service provider for verification and password reset messages
- The server file system for temporary staging of uploaded and generated files

2.1.2 User Interfaces

Web-based interfaces accessible through modern browsers:

- **Public:** Landing page, sign up page, and sign in page, with password reset flow
- **User:** Converter workspace with upload panel, parsing options, schema review table, preview and download controls, plus history and profile views
- **Administrator:** Dashboard for account management, activity monitoring, and configuration of limits
- **Common:** Navigation bar, session indicator, notification area, sign out

2.1.3 Hardware Interfaces

The system shall operate on standard web server hardware with no special hardware interface requirements. Client access is through standard computing devices with network connectivity.

2.1.4 Software Interfaces

- **Frontend:** React.js with routing and HTTP client libraries
- **Backend:** Python runtime with the FastAPI framework serving REST endpoints
- **Conversion Layer:** pandas, the Python `csv` module, and an ARFF library for header parsing and serialisation
- **Database:** PostgreSQL

2.1.5 Communications Interfaces

- HTTPS for all client-server communication
- RESTful API architecture as the frontend-backend interface
- JSON for metadata and errors; multipart form-data for uploads and binary download for converted files

2.1.6 Operations

- Automated daily database backups and regular health monitoring
- Automatic deletion of uploaded and generated files after the retention window
- Periodic security updates and patches

2.2 Product Functions

PF-1: User Management – Registration, sign in, session handling, password reset, and profile management

PF-2: File Upload and Format Detection – Accepting a dataset file, validating it, and determining whether the source format is CSV or ARFF

PF-3: Parsing and Schema Extraction – Reading CSV records per the configured delimiter and quoting rules, and reading ARFF relation, attribute, and data sections

PF-4: Attribute Type Inference – Classifying each column as numeric, nominal, date, or string and enumerating nominal value sets

PF-5: Schema Review and Override – Presenting the detected schema and accepting corrections before conversion

PF-6: Conversion – Generating a valid ARFF file from a parsed CSV dataset and a valid CSV file from a parsed ARFF dataset

PF-7: Validation and Error Reporting – Detecting malformed input and reporting the affected line and nature of each defect

PF-8: Preview and Download – Displaying the converted output and delivering the file to the user

PF-9: Conversion History – Recording each conversion and making it available for later reference

PF-10: Administration – Managing accounts, monitoring activity, and configuring system limits

2.3 User Characteristics

Primary Users – Students and Registered Users:

- Students preparing datasets for data mining and machine learning coursework
- Familiar with tabular data but not with the exact syntax of the ARFF header
- Access primarily via desktop and laptop browsers

Secondary Users – Researchers and Data Analysts:

- Technically proficient users moving datasets between analysis toolchains
- Require control over delimiters, inferred types, and missing-value representation
- Expect output accepted by WEKA without manual editing

Tertiary Users – Administrators:

- Project maintainers and laboratory staff, with high technical proficiency
- Responsible for account management, monitoring, and system upkeep

2.4 Constraints

- Must be compatible with Chrome, Firefox, Safari, and Edge
- Internet connectivity is required, as conversion is performed on the server
- Conversion is performed in memory, so input size is bounded by server memory and capped at a configured limit
- ARFF output must be accepted by WEKA 3.8 or later without modification, and CSV output must conform to RFC 4180
- Uploaded datasets must not be retained beyond the configured retention window
- Access is restricted to authenticated users, placing sign in on the critical path for every conversion

2.5 Assumptions and Dependencies

Assumptions:

- Stable internet connectivity and modern browsers with JavaScript enabled
- Uploaded files are plain text in UTF-8 or ASCII, neither compressed nor password protected
- CSV input is rectangular; every record has the same field count as the header row
- ARFF input follows the WEKA specification, with `@data` after all `@attribute` declarations
- Users have email access for verification and password reset

Dependencies:

- Availability of the chosen email service provider
- Continued availability of the third-party parsing libraries
- Hosting service availability for the frontend and backend deployments
- Stability of the published ARFF specification

3. Specific Requirements

3.1 Functional Requirements

FR-001: User Registration

- The system shall provide a sign up page allowing new users to create an account with name, email, and password
- The system shall verify email addresses through a confirmation link before activating the account
- The system shall enforce password strength rules and reject duplicate email addresses

FR-002: User Authentication

- The system shall provide a sign in page authenticating users through email and password
- The system shall issue a signed session token on success and reject requests to protected endpoints lacking a valid token
- The system shall provide password reset through a time-limited emailed link, and shall temporarily lock an account after repeated failed attempts

FR-003: Session Management

- The system shall expire a session after a configured period of inactivity
- The system shall provide a sign out control that invalidates the session token
- The system shall redirect unauthenticated users requesting the workspace to the sign in page

FR-004: User Profile Management

- The system shall allow users to view and edit profile information
- The system shall allow a password change after re-entering the current password
- The system shall allow account deletion, which also removes stored history

FR-005: File Upload

- The system shall allow authenticated users to upload a single `.csv` or `.arff` file
- The system shall reject files of unsupported extension, size, or encoding and state the reason

FR-006: Source Format Detection

- The system shall determine the source format from the file extension and, where ambiguous, from ARFF directives such as `@relation` in the leading lines
- The system shall display the detected source and target formats before conversion

FR-007: Parsing Configuration

- The system shall allow the user to set the delimiter, quote character, and whether the first CSV row is a header
- The system shall allow a relation name for the generated ARFF file, defaulting to the uploaded file name
- The system shall apply documented defaults when no selection is made

FR-008: CSV Parsing

- The system shall parse the CSV file into the internal representation, handling quoted fields, embedded delimiters, embedded newlines, and escaped quotes
- The system shall treat empty fields as missing values

FR-009: Attribute Type Inference

- The system shall classify each column as numeric, nominal, date, or string
- A column shall be numeric when all non-missing values parse as numbers, nominal when distinct values fall below a configured threshold, and string otherwise
- The system shall enumerate the value set of every nominal attribute

FR-010: Schema Review and Override

- The system shall display the detected schema as an editable table of attribute names and types
- The system shall allow the user to change any attribute type and rename attributes
- The system shall reject overrides inconsistent with the data and explain why

FR-011: CSV to ARFF Conversion

- The system shall generate a valid ARFF file with a `@relation` declaration, one `@attribute` declaration per column in original order, and a `@data` section holding one line per instance
- The system shall quote values containing spaces, commas, or quotes, and write missing values as the question mark symbol
- The generated file shall be accepted by WEKA without manual editing

FR-012: ARFF Parsing

- The system shall read the relation name, ordered attribute declarations with their declared types, and the instances of the data section
- The system shall ignore comment and blank lines anywhere in the file
- The system shall validate each value against the type declared for its attribute

FR-013: ARFF to CSV Conversion

- The system shall generate a CSV file whose first row holds attribute names and whose subsequent rows each hold one instance
- The system shall quote and escape fields containing the delimiter, quotes, or newlines per RFC 4180, and write ARFF missing values as empty fields

FR-014: Input Validation and Error Reporting

- The system shall detect inconsistent field counts, unterminated quoted fields, ARFF files with no @data section, type violations, and duplicate attribute names
- Each report shall identify the affected line number and describe the defect
- The system shall abort on a fatal error and never deliver a partial output file

FR-015: Output Preview

- The system shall preview the converted output, showing the full generated ARFF header or the first rows of the generated CSV
- The system shall display the instance and attribute counts alongside the preview

FR-016: Download Converted File

- The system shall allow download of the converted file, named from the uploaded file
- The download shall be served over the same authenticated session

FR-017: Conversion History

- The system shall record each conversion with timestamp, source and target format, file name and size, instance and attribute counts, and status
- The system shall display history to the owning user in reverse chronological order and allow deletion of individual entries

FR-018: Administrative Dashboard

- The system shall provide an admin panel for viewing accounts, activating or suspending them, and viewing aggregate statistics
- The system shall allow configuration of the maximum upload size and the nominal inference threshold
- The system shall restrict administrative endpoints to the administrator role

3.2 Design Constraints

C-001: Technology Stack

- The frontend shall use React.js; the backend shall use Python with FastAPI
- The frontend-backend interface shall follow a RESTful API architecture
- PostgreSQL shall be the database management system

C-002: Separation of Concerns

- The frontend shall contain no conversion logic; all parsing, inference, and serialisation shall occur in the backend
- Readers, writers, and the conversion engine shall be separate modules communicating only through the internal representation

C-003: Format Fidelity

- ARFF output shall conform to the WEKA specification and CSV output to RFC 4180
- Conflicts between user options and these specifications shall be resolved in favour of the specification

C-004: Database Design

- The system shall implement a normalised database schema
- Data integrity shall be maintained through constraints, foreign keys, and server-side validation

C-005: Security Standards

- HTTPS shall be used for all communications, with secure authentication and session management
- Uploaded files shall be treated as untrusted input and parsed so that their contents can never be executed

3.3 Non-Functional Requirements

3.3.1 Performance Requirements

NF-001: Response Time

- Web pages shall load within 3 seconds under normal conditions
- Upload of a file up to 10 MB shall be acknowledged within 2 seconds
- Conversion of a file up to 10 MB or 100,000 instances shall complete within 10 seconds on the reference configuration

NF-002: Capacity

- The system shall support up to 200 concurrent authenticated users
- The system shall accept files up to 50 MB and process up to 2000 conversions per day

3.3.2 Reliability Requirements

NF-003: System Reliability

- The system shall maintain 99% uptime during operating hours and recover from failures within 2 hours
- Malformed input shall never terminate the application abnormally; every parse failure shall be surfaced as a descriptive message
- Conversion shall be atomic, leaving no partial file and no partial history record

3.3.3 Availability Requirements

NF-004: System Availability

- The system shall be continuously available for authenticated user access
- Administrative functions shall be available during business hours
- Maintenance windows shall be scheduled during low-usage periods and announced

3.3.4 Security Requirements

NF-005: Authentication Security

- Passwords shall be stored using a salted industry-standard hash and never logged in plain text
- Session tokens shall expire after 24 hours of inactivity
- Access to administrative data shall be role-based and logged

NF-006: Data Protection

- Uploaded datasets shall be accessible only to the uploading user and deleted within the configured retention window
- Uploaded content shall not be used for any purpose other than the requested conversion
- The system shall protect against injection, cross-site scripting, and unrestricted file upload

3.3.5 Maintainability Requirements

NF-007: Code Quality

- Code shall follow PEP 8 for Python and the adopted JavaScript style guide with React best practices
- Each reader and writer shall be unit tested, including a CSV-to-ARFF-to-CSV round-trip test
- Size limits and thresholds shall be externalised rather than hard-coded

3.3.6 Portability Requirements

NF-008: Platform Independence

- The backend shall run on Windows, Linux, and macOS with Python 3.8 or higher
- The frontend shall function on all supported browsers without platform-specific code
- The database connection shall be configurable via environment variables

3.3.7 Correctness Requirements

NF-009: Data Integrity of Conversion

- Conversion shall be lossless in instance count, attribute count, attribute order, and every non-missing value
- A CSV converted to ARFF and back shall equal the original up to quoting and whitespace, with numeric values written without loss of precision

3.4 External Interface Requirements

3.4.1 User Interface Requirements

UI-001: General Interface Standards

- The interface shall follow modern web design principles and remain consistent across pages
- Navigation shall clearly separate public pages from the authenticated workspace
- Errors and warnings shall appear in a dedicated notification area beside the control that caused them

Input Screens:

- Sign up form with name, email, password, and confirm password fields
- Sign in form with email and password fields and a forgot password link
- Upload panel with file selector and drag-and-drop area
- Parsing options panel with delimiter, quote character, header row, and relation name
- Schema review table with an editable type selector per attribute

Output Screens:

- Detected format banner showing source and target formats
- Data preview grid and converted output preview with instance and attribute counts
- Conversion history view with per-entry status
- Admin dashboard showing accounts, activity, and configured limits

3.4.2 Hardware Interface Requirements

HI-001: Client Hardware

- The system shall operate on standard computing devices running a modern browser
- Standard network connectivity is required for access
- The server shall require at least 4 GB RAM and disk space to stage uploads

Input Requirements: Standard keyboard and mouse for desktop users; local file system access for selecting the dataset to upload.

Output Requirements: Standard display monitors for the workspace and preview grids; local file system write access for downloading the converted file.

3.4.3 Software Interface Requirements

SI-001: Conversion Library Integration

- The backend shall integrate with pandas, the Python `csv` module, and an ARFF library for parsing and serialisation
- Malformed input shall raise a structured error rather than propagating a library exception to the client

Input Data Format: Uploaded file bytes with a parsing configuration object in JSON.

Output Data Format: Converted file bytes with a schema summary and warning list in JSON.

SI-002: Database Integration

- The backend shall interface with PostgreSQL through an object-relational mapping layer
- Credentials, profile data, roles, and conversion history shall be persisted; dataset contents shall not be

Data Formats: Normalised user tables with a unique constraint on email; history records with a foreign key to the owning user; timestamps in ISO 8601 format.

SI-003: Email Service Integration

- The system shall integrate with an email provider for verification and password reset messages
- The integration shall handle delivery confirmation and failure without blocking registration

Input Requirements: SMTP configuration, HTML templates, recipient addresses.

Output Requirements: Verification and reset links, delivery status, sending logs.

3.4.4 Communications Interface Requirements

CI-001: API Standards

- All endpoints shall follow RESTful design principles and shall be versioned
- Responses shall use standard HTTP status codes, returning 400 for malformed input, 401 for invalid authentication, and 413 for oversized uploads
- Protected endpoints shall require a bearer token issued at sign in

CI-002: Data Formats

- Metadata, parsing options, schema descriptions, and errors shall use JSON
- File uploads shall use multipart form-data; dates shall follow ISO 8601

CI-003: Network Communications

- All client-server communication shall use HTTPS
- Cross-origin requests shall be restricted to the registered frontend origin
- API rate limiting shall prevent abuse, with timeouts configured for large uploads

4. Appendix A: Protocols and Standards

Data Format Standards:

- **RFC 4180 (CSV)** – Defines record separation, field quoting, escaping of embedded quotes, and the optional header line; conformance ensures generated CSV opens correctly in standard spreadsheet and analysis software.
- **ARFF Specification** – Defines a header declaring a relation name and typed attributes, followed by a data section of one instance per line; numeric, nominal, string, and date types are supported and a question mark denotes a missing value.
- **ISO 8601** – The standard for representing dates and times, used for ARFF date attributes and all history timestamps.
- **UTF-8** – The encoding assumed for all input and produced for all output, so that non-ASCII values survive conversion.

Web Protocols:

- **HTTP/HTTPS** – Standard protocols for transferring data, with HTTPS providing TLS-encrypted communication protecting credentials and dataset contents in transit.
- **RESTful API standards** – An architectural style using standard HTTP methods for stateless communication, forming the interface between the React frontend and the Python backend.
- **JSON** – A lightweight text-based interchange format used for parsing options, schema descriptions, and error responses across the API.
- **multipart/form-data** – The encoding used to transmit the uploaded file with its parsing parameters in a single request.

Authentication Standards:

- **JWT (JSON Web Tokens)** – Digitally signed JSON objects used to maintain sessions and authenticate API requests without server-side session storage.
- **Password hashing (bcrypt)** – A hash designed for password storage, with salt generation and multiple rounds resisting rainbow table and brute force attacks.
- **Session management protocols** – Secure token handling, expiry after inactivity, and explicit invalidation on sign out.

Security Standards:

- **TLS encryption** – Protects all data between browser and server, including credentials and uploaded datasets, from interception.
- **Input validation and sanitisation** – File contents are treated as untrusted data that is parsed but never executed or interpolated into queries.
- **File upload safety controls** – Extension and content-type checks, size limits, randomised temporary names, and storage outside the web root.

Communication Standards:

- **SMTP** – The standard protocol for sending email, used to deliver account verification links and password reset messages.

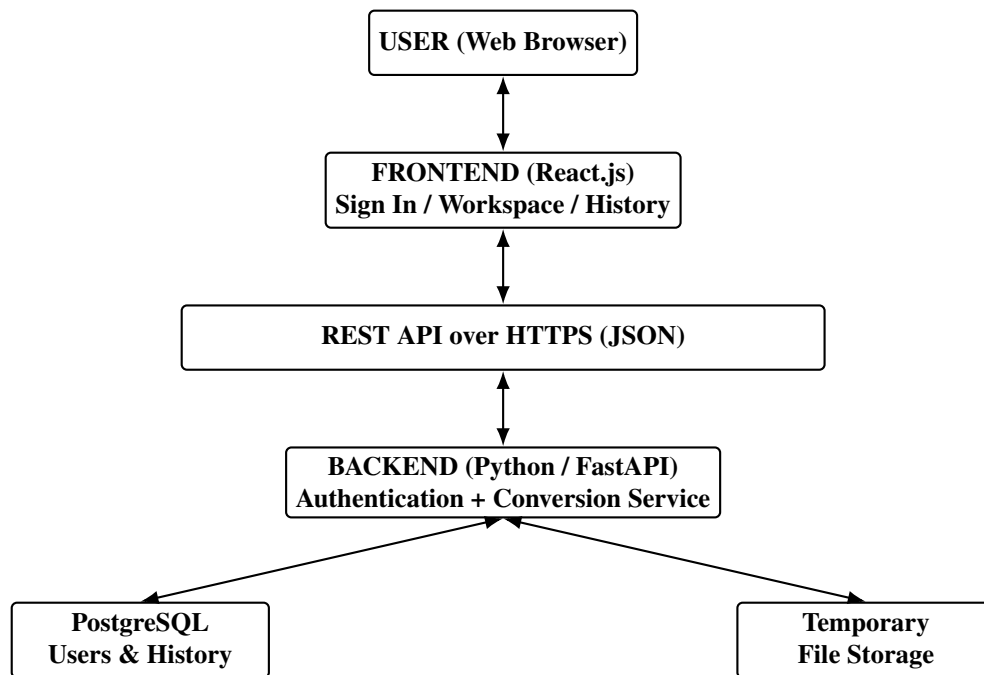
Database Standards:

- **SQL standards** – Ensure consistent database operations, retrieval, and transaction management.
- **ACID properties** – Guarantee reliable transactions, so a history record is written only when a conversion completes successfully.
- **Referential integrity constraints** – Ensure every history record refers to an existing user account.

5. Appendix B: System Architecture

The diagram shows the overall structure of the system and the interface between the frontend and the backend. The user interacts only with the React frontend, which communicates with the Python backend exclusively through the REST API over HTTPS. The backend authenticates every request before passing

it to the conversion service, which parses the uploaded file, applies the confirmed schema, and writes the output in the target format. Accounts and conversion history are held in the database, while uploaded and generated files are kept in temporary storage and deleted after the retention window.



6. Index

A	Administrative Dashboard, 6	Missing Value, 5
	ARFF (Attribute-Relation File Format), 1	N
	Authentication, 4	Nominal Attribute, 5
C		P
	Conversion Engine, 3	PostgreSQL, 3
	Conversion History, 6	
	CSV (Comma-Separated Values), 1	R
D		React.js, 3
	Delimiter, 5	REST / RESTful, 3
E		S
	Error Reporting, 6	Schema, 5
F		Sign In / Sign Up Page, 2
	File Upload, 5	SRS (Software Requirements Specification), 1
H		T
	HTTPS, 3	Type Inference, 3
J		U
	JSON (JavaScript Object Notation), 3	User Interfaces, 2
M		User Management, 3
		User Registration, 4

Design Document

for

File Converter System

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1 Data Flow Diagram Model

1.1 Level-0: Context Diagram

The Level 0 DFD represents the File Converter System as a single process and shows its interaction with the external entities: the Registered User, the Administrator, and the Email Service.

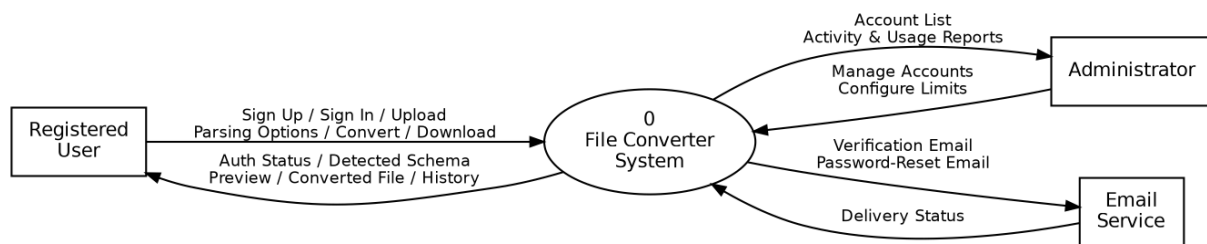
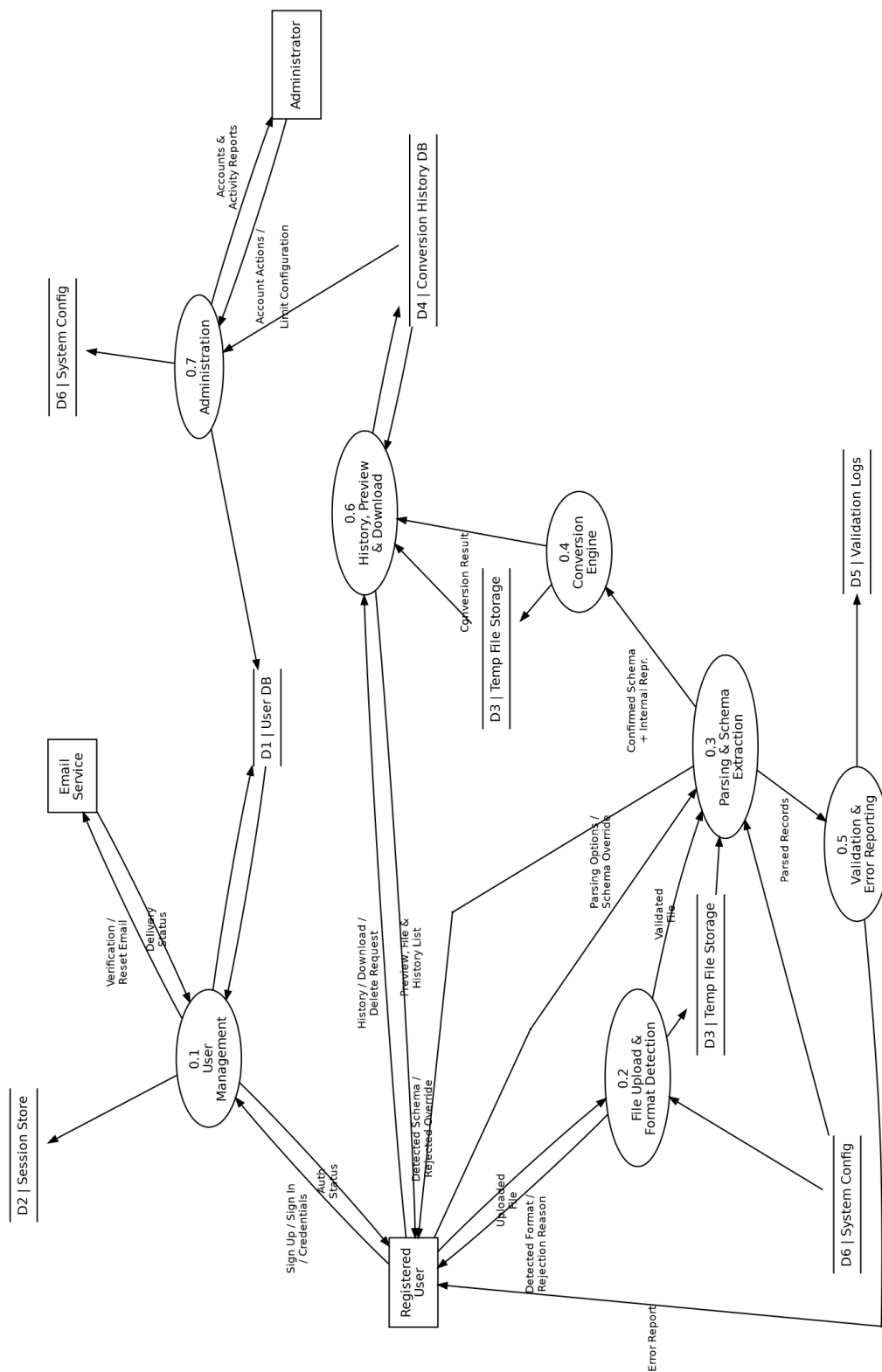


Figure 1: Level 0 Context Diagram

1.2 Level-1: Top-Level Decomposition

The Level 1 DFD decomposes the system into its seven major sub-processes — User Management, File Upload & Format Detection, Parsing & Schema Extraction, Conversion Engine, Validation & Error Reporting, History/Preview/Download, and Administration — together with the data stores each one reads from or writes to. It is shown rotated on the following page so that every flow label stays legible; the D3 Temp File Storage and D6 System Config stores are drawn more than once to keep the arrows short and free of crossings.



1.3 Level-2 DFDs

1.3.1 User Management (Process 0.1)

Expands registration, email verification, login, session (JWT) management, password reset, and profile management, mapping to FR-001 through FR-004 of the SRS.

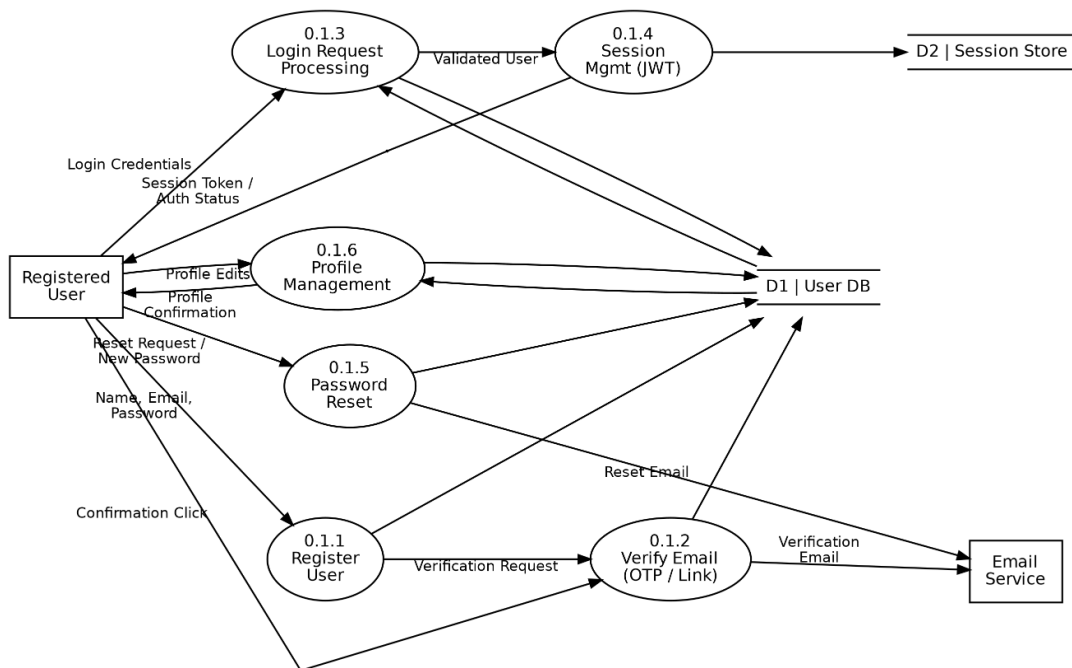


Figure 3: Level 2 DFD for 0.1 User Management

1.3.2 File Upload & Format Detection (Process 0.2)

Expands upload validation, source-format detection, and the detected-format banner, mapping to FR-005 and FR-006.

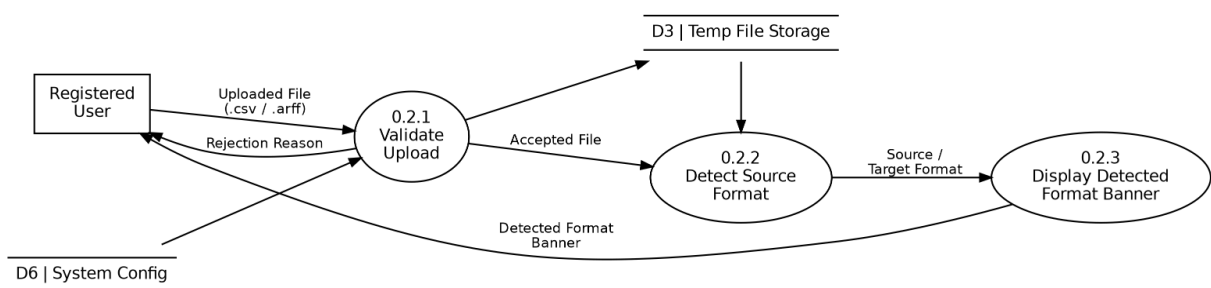


Figure 4: Level 2 DFD for 0.2 File Upload & Format Detection

1.3.3 Parsing & Schema Extraction (Process 0.3)

Expands CSV parsing, ARFF parsing, attribute type inference, and schema review with user override, mapping to FR-007 through FR-010.

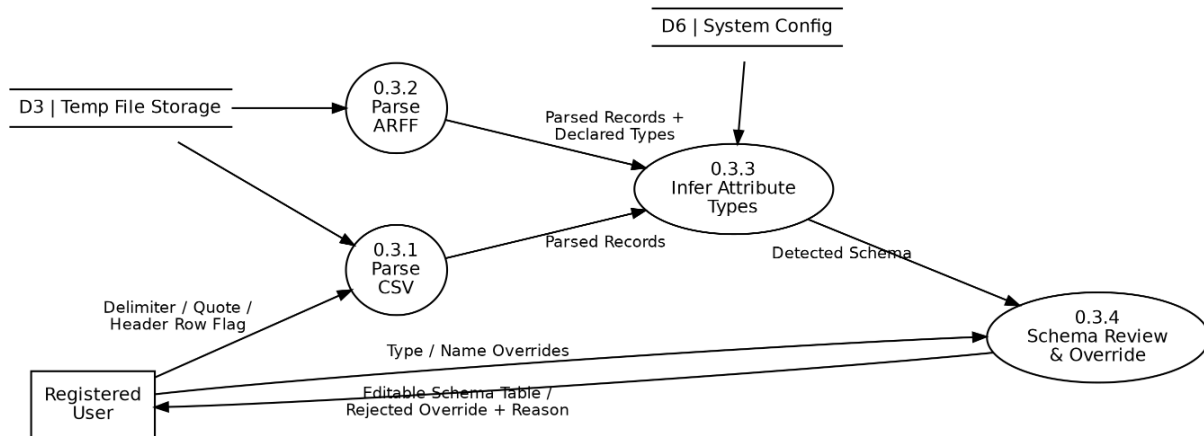


Figure 5: Level 2 DFD for 0.3 Parsing & Schema Extraction

1.3.4 Conversion Engine (Process 0.4)

Expands the format-independent internal representation and the two conversion directions, mapping to FR-011 through FR-013.

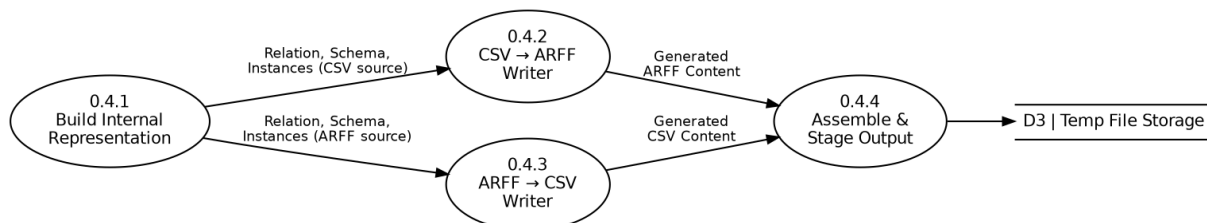


Figure 6: Level 2 DFD for 0.4 Conversion Engine

1.3.5 Validation & Error Reporting (Process 0.5)

Expands structural validation, schema type validation, error-report compilation, and abort-on-fatal-error handling, mapping to FR-014.

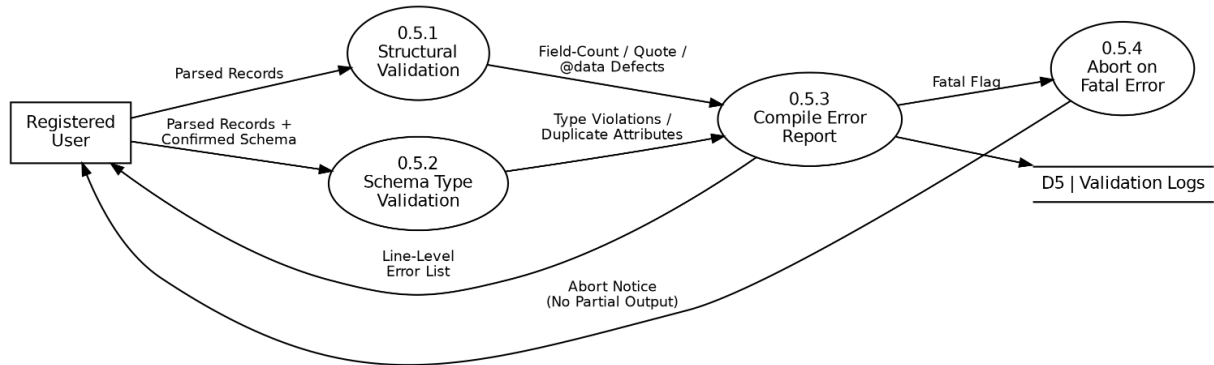


Figure 7: Level 2 DFD for 0.5 Validation & Error Reporting

1.3.6 History, Preview & Download (Process 0.6)

Expands output preview, file download, and conversion-history recording, listing, and deletion, mapping to FR-015 through FR-017.

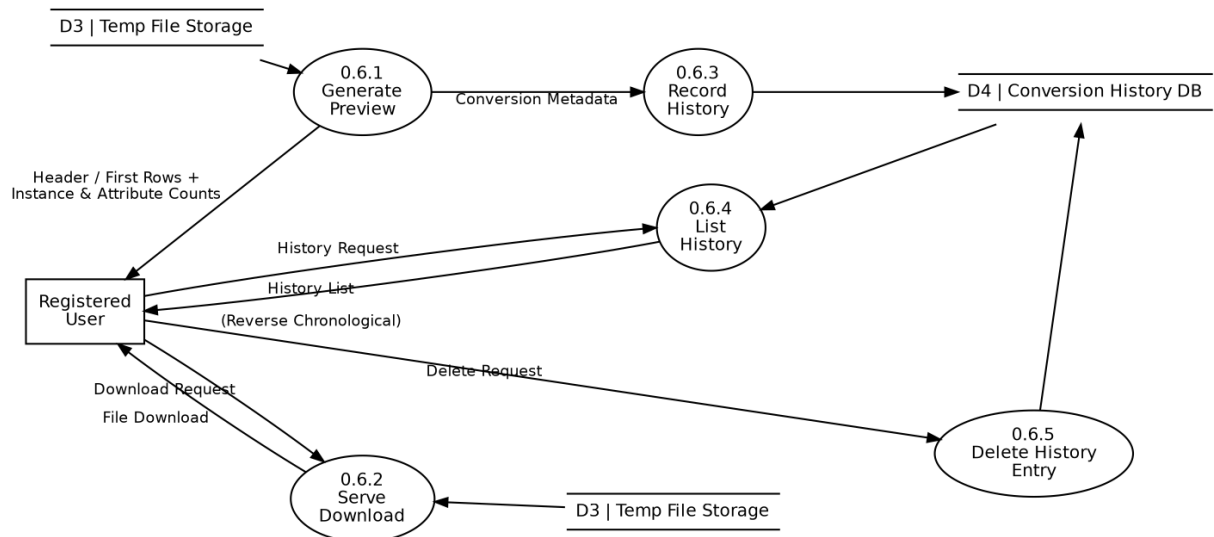


Figure 8: Level 2 DFD for 0.6 History, Preview & Download

1.3.7 Administration (Process 0.7)

Expands account management, activity monitoring, and configuration of system limits, mapping to FR-018.

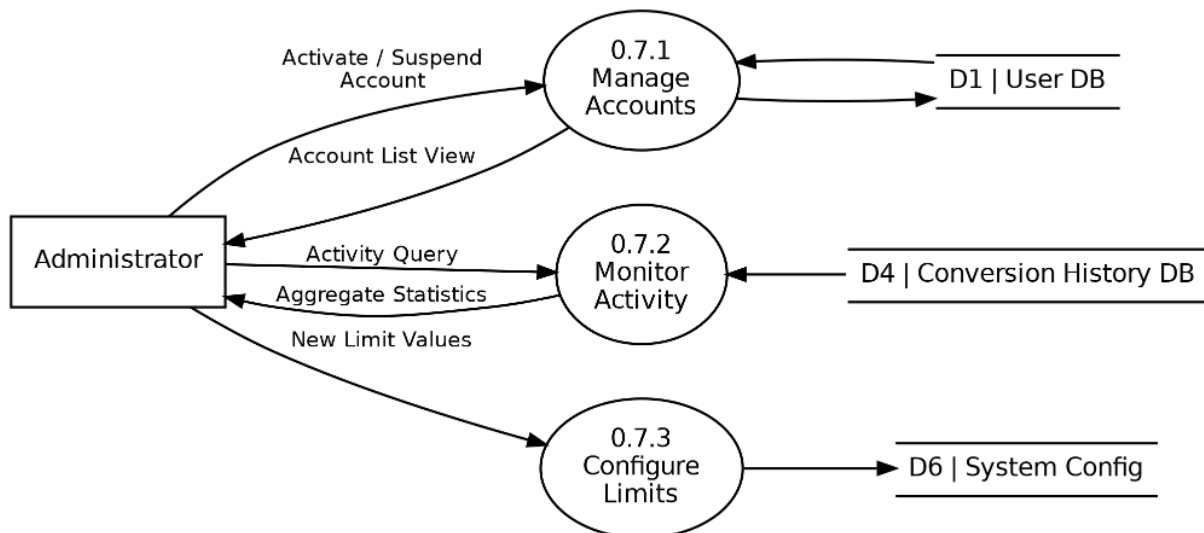


Figure 9: Level 2 DFD for 0.7 Administration

1.4 Level-3 DFDs

1.4.1 Session Management, JWT Generation (Process 0.1.4)

Expands JWT issuance: payload assembly, signing, refresh-token generation, and storage of token metadata for the session issued at login.

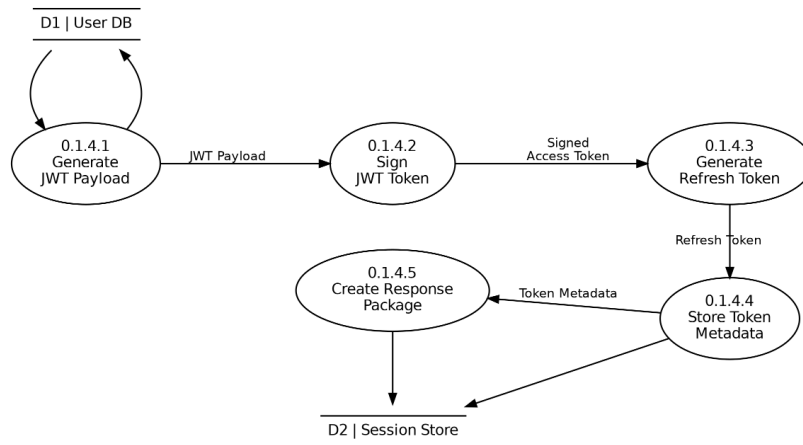


Figure 10: Level 3 DFD — Session Management (JWT Generation)

1.4.2 Attribute Type Inference (Process 0.3.3)

Expands the per-column classification logic: numeric parse attempt, distinct-value counting against the configured nominal threshold, and enumeration of nominal value sets.

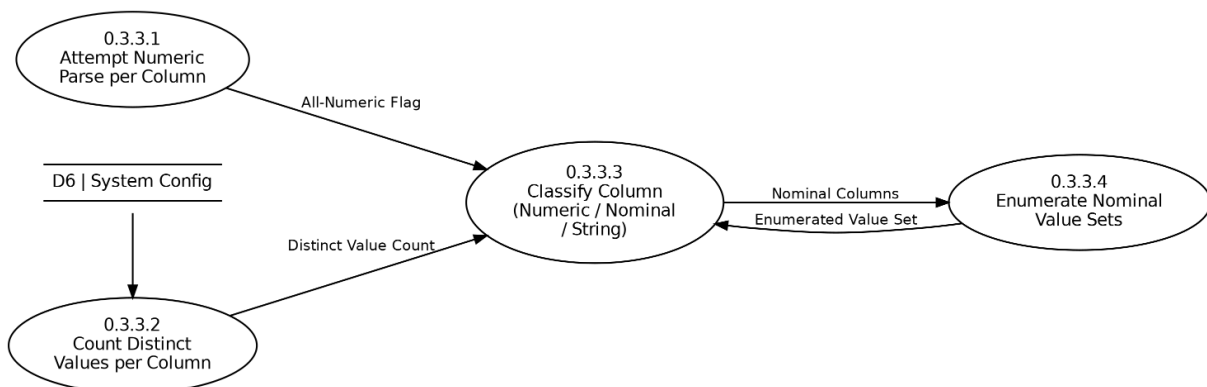


Figure 11: Level 3 DFD — Attribute Type Inference

1.4.3 CSV to ARFF Writer (Process 0.4.2)

Expands ARFF generation: writing the @relation declaration, the @attribute declarations in original column order, and the @data section with quoting and missing-value encoding.

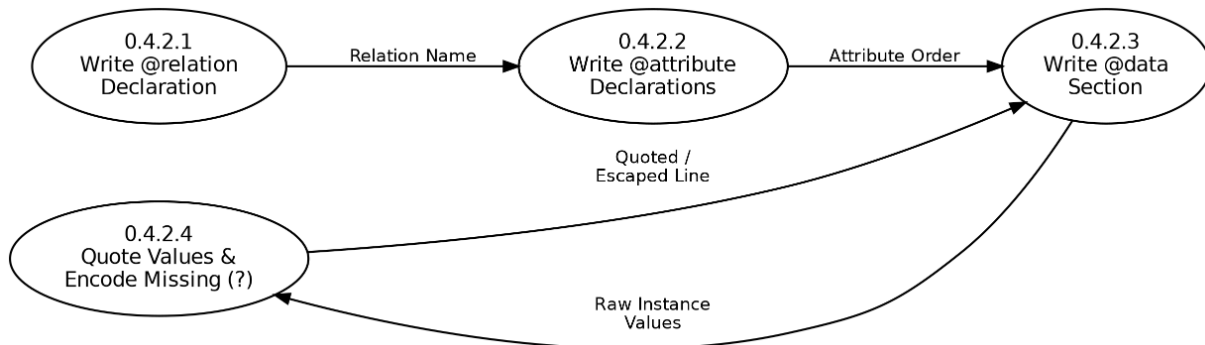


Figure 12: Level 3 DFD — CSV to ARFF Writer

1.4.4 Compile Error Report (Process 0.5.3)

Expands the compilation of validation defects into the final report delivered to the user, including fatal/non-fatal classification.

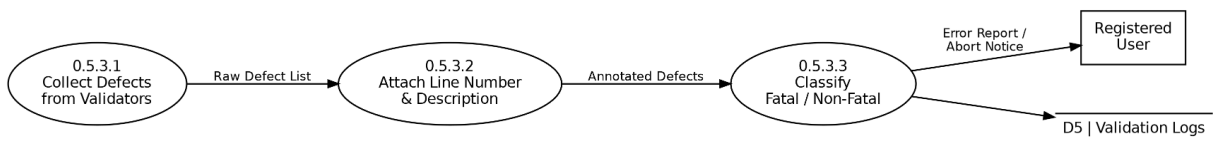


Figure 13: Level 3 DFD — Compile Error Report

Structural Chart

for

File Converter System (CSV to ARFF and ARFF to CSV)

Team Members:

Deepanshu Kumar (241IT020)

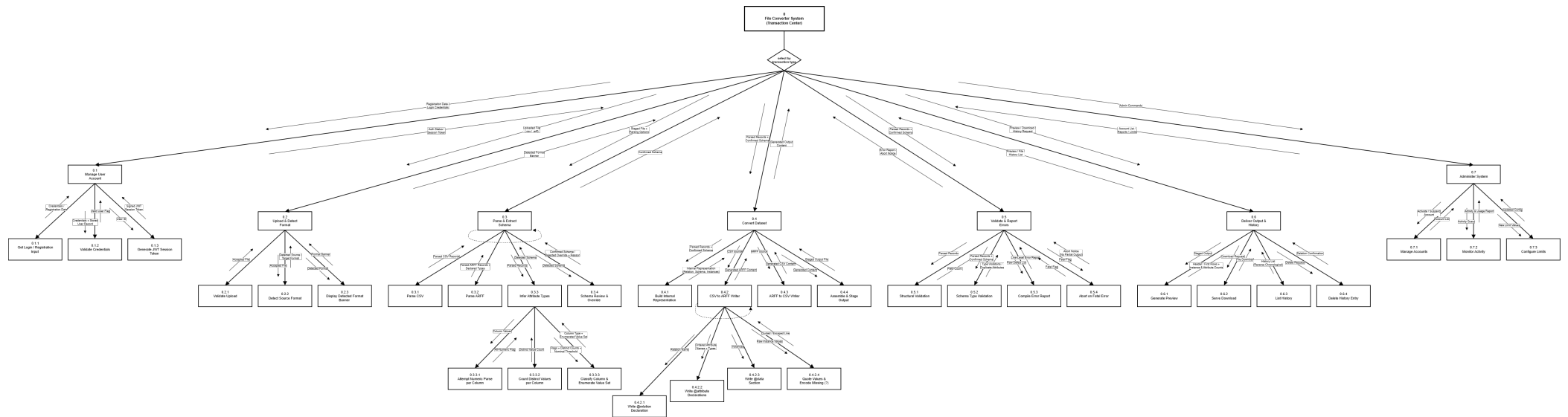
Aayush Sarraf (241IT001)

Dhruv Agarwal (241IT026)

Aditya Raj (241IT006)

Department of Information Technology, NITK

Figure 14: Structural Chart



4. Data Dictionary

4.1 External Entities

Entity Name	Description
Registered User	Users (such as students or researchers) who authenticate to upload datasets, configure parsing, and download converted ARFF or CSV files.
Administrator	System maintainers with high technical proficiency responsible for managing user accounts, monitoring activity, and configuring system limits.
Relational DBMS	The PostgreSQL database used to store user accounts, active sessions, and conversion histories.
Email Service Provider	An external service integrated to dispatch account verification links and password recovery messages.
WEKA Workbench	The external machine learning environment that dictates the required syntax and formatting for the generated ARFF outputs.

4.2 Key Data Flows

User Management System

Data Flow	Data Elements	Type	Description
Registration Data	Name, Email, Password	Structured	Information submitted by a new user to create an account.
Login Credentials	Email, Password	Structured	User inputs required for authentication on the sign-in page.
Session Token	JWT_String, Expiry_Time	Structured	A signed JSON Web Token issued upon successful authentication to manage stateless sessions.
Profile Data	Name, Email, Password_Hash	Structured	User information retrieved or updated through the profile management interface.

File Upload & Parsing Configuration

Data Flow	Data Elements	Type	Description
Uploaded Dataset	File_Bytes, File_Name, File_Extension, Encoding	Binary/Text	The source .csv or .arff file submitted by the user.
Format Metadata	Source_Format, Target_Format	Structured	The automatically detected format of the uploaded file and its intended conversion target.
Parsing Options	Delimiter, Quote_Character, Header_Row_Boolean, Relation_Name	Structured	User-defined settings that dictate how the CSV or ARFF file should be parsed.

Schema Extraction & Review

Data Flow	Data Elements	Type	Description
Detected Schema	Attribute_Names, Attribute_Types, Nominal_Value_Sets	Structured	The ordered list of columns and their inferred data types extracted from the parsed file.
Schema Overrides	Modified_Attribute_Name, Modified_Attribute_Type	Structured	User-submitted corrections to the automatically detected schema prior to conversion.
Internal Representation	Relation_Name, Ordered_Attribute_Schema, Instances_Collection	Structured	The format-independent core data structure used in memory during bidirectional conversion.

Conversion & Validation

Data Flow	Data Elements	Type	Description
Validation Error Report	Affected_Line_Number, Defect_Nature	Structured	Detailed report of malformed input (e.g., inconsistent field counts, type violations) preventing conversion.
Converted Output	Output_File_Bytes, Generated_File_Name	Text	The final generated valid ARFF or CSV file ready for the user to download.
Output Preview	Generated_Header, Preview_Rows, Instance_Count, Attribute_Count	Structured	A visual snippet of the converted dataset displayed on the UI prior to download.

History & Administration

Data Flow	Data Elements	Type	Description
Conversion Record	Timestamp, Source_Format, Target_Format, File_Name, Status	Structured	Details recorded for each individual conversion action, stored for user reference.
System Limits	Max_Upload_Size, Nominal_Inference_Threshold	Structured	Administrator-configured constraints that govern system performance and behavior.
Account Actions	User_Email, Account_Status	Structured	Commands issued by the administrator to activate, suspend, or delete user accounts.